OmniSpace OS — Week 1 Research Report

Spatial UI Placement for AR/VR Environments

1. Introduction

OmniSpace OS is a project focused on designing a spatial user interface for Augmented Reality (AR) and Virtual Reality (VR) environments. Unlike traditional operating systems that display applications on a flat screen, spatial interfaces allow digital windows, tools, and information panels to be arranged within a three-dimensional environment.

The primary focus of Week 1 is to research spatial UI placement and understand how interface elements can be organized according to their importance, accessibility, and user comfort.

2. Problem Statement

Traditional user interfaces are designed for two-dimensional screens and may not be suitable for spatial computing environments without modification. Poorly positioned floating panels can create visual clutter, require unnecessary head movements, and make information difficult to access.

OmniSpace OS aims to explore a structured UI placement approach that improves interface organization, accessibility, and comfort in spatial environments.

3. Objectives

- To understand the fundamental concepts of spatial computing and spatial user interfaces.
- To study the importance of interface placement in AR/VR environments.
- To categorize interface elements according to their importance and frequency of use.
- To develop an initial UI placement layout using Figma.
- To establish a foundation for future 3D prototyping and usability evaluation.

4. Research Background

Spatial computing enables users to interact with digital content within a three-dimensional environment. It introduces new design considerations, including viewing distance, viewing angle, visual hierarchy, reachability, and the amount of physical movement required to interact with interface elements.

An effective spatial interface should prioritize important information, provide easy access to supporting tools, and minimize distractions caused by unnecessary visual elements.

The interface should also consider readability, accessibility, consistency, and individual user comfort.

5. Spatial UI Placement Zones

For the initial OmniSpace design, the interface is organized into three conceptual zones.

5.1 Primary Zone

The Primary Zone contains the most important interface elements and frequently used functions. These elements should be easy to locate and view.

Examples:

- Main Workspace
- Application Launcher
- Frequently Used Controls
- Quick Access Functions

5.2 Secondary Zone

The Secondary Zone contains supporting tools and information that users access when required. These elements should remain accessible without competing with the main workspace.

Examples:

- Files
- Notes
- Calendar
- Settings
- Information Panels

5.3 Peripheral Zone

The Peripheral Zone contains low-priority information that users may need to check occasionally. These elements should remain subtle and avoid interrupting the user's primary task.

Examples:

- System Status
- Connection Status
- Time
- Non-urgent Notifications

These three zones represent the proposed organizational framework for OmniSpace OS. Their effectiveness will need to be evaluated through further design iterations and user feedback.

6. Design Principles

The following principles guide the initial UI placement design.

6.1 Visual Hierarchy: Important interface elements should receive greater visual emphasis than secondary information.

6.2 User Comfort: Frequently used elements should be positioned to reduce unnecessary head movements and uncomfortable reaching.

6.3 Accessibility: Text, controls, and information should be easy to identify and interact with.

6.4 Consistency: Interface elements should follow consistent spacing, alignment, typography, and design patterns.

6.5 Reduced Visual Clutter: The interface should display only the information necessary for the user's current task.

6.6 User Control: Where appropriate, users should be able to reposition or customize interface panels according to their preferences.

7. Week 1 Methodology

The initial research and design process consists of the following activities:

1. Literature Review: Study spatial UI concepts and introductory AR/VR design guidelines.
2. Requirement Identification: Identify the interface elements required for the OmniSpace dashboard.
3. Content Prioritization: Classify interface elements into primary, secondary, and peripheral categories.
4. Wireframe Planning: Determine the initial placement of the Main Workspace, Tools panel, Information panel, and System Status panel.
5. Low-Fidelity Design: Use Figma to represent the proposed layout through simple rectangles, labels, and interface components.
6. Review and Refinement: Identify areas for improvement and prepare the design for further development.

8. Proposed UI Placement Layout

The initial OmniSpace dashboard is planned with the following components:

Component	Proposed placement	Purpose
Main Workspace	Centre	Displays the main working area and frequently used functions
Tools Panel	Left	Provides access to files, notes, calendar, and settings
Information Panel	Right	Displays workspace information and recent files
System Status Panel	Bottom or peripheral area	Displays connection status, time, and notifications

The proposed layout establishes a clear relationship between the interface components. The Main Workspace receives the highest visual priority, while supporting tools and system information remain accessible without dominating the interface.

The initial wireframe is a two-dimensional representation of the proposed spatial interface and does not simulate actual three-dimensional distances or field of view.

9. Tools and Technologies

Figma: Used to create the initial UI placement layout, wireframes, and interface components. It helps visualize the proposed structure before moving towards 3D prototyping.

Spline: A potential tool for the next stage of development, allowing the team to explore three-dimensional interface panels and spatial visualizations.

10. Expected Week 1 Deliverables

The expected outcomes of the first week are:

- A research summary on spatial UI placement.
- A defined problem statement and project objectives.
- A classification of interface elements into three conceptual zones.
- An initial low-fidelity UI placement wireframe in Figma.
- A list of design considerations and questions for future evaluation.

11. Limitations

The initial design is conceptual and has not yet established validated panel distances, viewing angles, or interaction methods. These factors depend on the target device, application requirements, and individual user needs.

The proposed placement zones should therefore be treated as an initial design framework rather than a universally applicable spatial UI standard.

12. Future Work

The next stage of the project may include:

- Refining the Figma wireframe based on feedback.
- Improving panel spacing, alignment, and visual hierarchy.
- Exploring a 3D interface prototype using Spline.
- Evaluating the usability and accessibility of the proposed layout.
- Investigating suitable interaction methods for spatial environments, depending on the project scope.

13. Conclusion

Week 1 establishes the research foundation for OmniSpace OS by identifying the challenges associated with spatial UI placement and proposing a structured approach to interface organization.

The primary, secondary, and peripheral zones provide an initial framework for arranging important controls, supporting tools, and system information. The proposed Figma wireframe will serve as the foundation for further interface refinement, prototyping, and usability evaluation.

14. References

1. Apple Developer — Human Interface Guidelines. https://developer.apple.com/design/human-interface-guidelines/
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3. Nielsen Norman Group — Virtual Reality User Experience. https://www.nngroup.com/topic/virtual-reality/
4. ISO 9241-210:2019 — Ergonomics of Human-System Interaction: Human-Centred Design for Interactive Systems. https://www.iso.org/standard/77520.html